

Question Paper

Exam Date & Time: 11-Nov-2019 (10:00 AM - 01:00 PM)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Science in Information Technology (B.Sc. IT)

Semester-I University Examination November-2019

Date: 11.11.2019, Monday

Time: 10.00 a.m. to 01.00 p.m.

INTRODUCTION TO PROGRAMMING IN C [CA402]

Marks: 70

Duration: 180 mins.

Section-I

Answer all the questions.

- 1) Processing step in a flowchart is represented by _____. (1)
[Parallelogram](#) [Arrow](#) [Diamond](#) [Rectangle](#)
- 2) _____ means that each step of the algorithm is executed in the specified order. (1)
[Decision](#) [Sequence](#) [Repetition](#) [None of the above](#)
- 3) All the preprocessor commands start with _____ symbol. (1)
[/*](#) [//](#) <#> [{](#)
- 4) The compiler translates object code into source code. (1)
[True](#) [False](#)
- 5) What will be the value of y and x? (1)
x = 5;
y = x++;
[y=6, x=6](#) [y=5, x=6](#) [y=6, x=5](#) [y=5, x=5](#)
- 6) _____ is non-executable statement. (1)
[Comment](#) [Variable](#) [Array](#) [Conditional statement](#)
- 7) _____ symbol is used as an assignment operator. (1)
[==](#) [&](#) <#> [≡](#)
- 8) The default must be placed at the end in switch case. (1)
[True](#) [False](#)
- 9) C has a built-in multi way decision statement known as _____. (1)
[if](#) [while](#) [for](#) [switch](#)
- 10) In _____ statement, the body of the loop is always executed atleast once. (1)
[while](#) [do](#) [for](#) [none of the above](#)
- 11) ANSI C allows upto _____ level of nesting (1)
[32](#) [256](#) [52](#) [15](#)
- 12) An array index always begins with _____. (1)
[0](#) [1](#) [-1](#) [size-1](#)
- 13) What will be the value of rollno[3]=_____. (1)
int rollno[4]={1,2,3};
[3](#) [Null character](#) [0](#) [2](#)
- 14) The character string is terminated with _____. (1)
[/0](#) [\0](#) [0](#) [-1](#)
- 15) An array can store different data of the same type. (1)
[True](#) [False](#)
- 16) All the string functions are available in _____ header file. (1)
[stdio.h](#) [math.h](#) [conio.h](#) [string.h](#)
- 17) _____ function joins two strings together. (1)
[strcpy](#) [strcmp](#) [strlen](#) [strcat](#)
- 18) The return type is optional, when the function returns _____ type data. (1)

[void](#) [int](#) [float](#) [char](#)

- 19) Actual parameters are available in _____. (1)
[function call](#) [function definition](#) [function prototype](#) [function declaration](#)
- 20) A function may have more than one return statement. (1)
[True](#) [False](#)

Section-II

Answer all the questions.

- 1) Explain all the symbols used in flowchart. (6)
- A) (6)
[OR] Describe the characteristics of C programming language. (6)
B) (6)
C) Explain all the operators available in C. (6)
[OR] Write down rules of switch statement. (6)
D)
- 2) Explain with diagram structure of C program. (5)
- A) (5)
[OR] What is called identifier? Write down rules of identifiers (5)
B)
- 3) Define Algorithm. Explain key features of algorithm. (4)
- A) (4)
[OR] Draw a flowchart to find largest of three numbers. (4)
B) (4)
C) Explain files used in C program. (4)
[OR] Write an algorithm to find sum of first N natural numbers. (4)
D)

Section-III

Answer all the questions.

- 1) Explain following functions in detail with example: (6)
1. strcat()
A) 2. strcmp()
3. strcpy() (6)
[OR] Explain function definition with all elements and example. (6)
B) (6)
C) Describe categories of functions in details with example. (6)
[OR] What is called constant? Explain types of constants. (6)
D)
- 2) Explain with diagram compilation and execution of C program. (5)
- A) (5)
[OR] Define Array. Explain declaration and initialization of one dimensional array. (5)
B)
- 3) What is called loop? Explain types of loop with example. (4)
- A) (4)
[OR] Give difference between compiler and interpreter. (4)
B) (4)
C) Write a C program to display average of N numbers entered by user using an array. (4)
[OR] Write a C program to find square of a number using function. (4)
D)

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